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10 / 10 submitted

Q1. Peak Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
a = [1,3,20,4,1,0]
for i in range(1,len(a)-1):
    if a[i]>a[i-1] and a[i]>a[i+1]:
        print(a[i])
        break
```

OUTPUT

20

Q2. Second Largest and Second Smallest - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
n = 6
arr=[10,20,5,40,15,30]
t=tuple(sorted(set(arr)))
if len(t)<2:
    print("second smallest and second largest do not exist.")
else:
    print(t[1])
    print(t[-2])
```

INPUT

6

OUTPUT

10
30**Q3. Frequency of Each Element - List**

Score: 0.00 TC: 0/1 passed

CODE

```
arr =[1,2,3,2,4,1,5,2]
freq={}
for i in arr:
    if i in freq:
        freq[i]+=1
    else:
        freq[i]=1
for key in sorted(freq):
    print(f"{key}={freq[key]}")
```

INPUT

1 2 3 2 4 1 5 2

OUTPUT

1=2
2=3
3=1
4=1
5=1

Q4. Missing Number - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
n = 5
t=(1,2,3,5,6)
total = (n+1)*(n+2)//2
missing=total-sum(t)
print(missing)
```

OUTPUT

4

Q5. Majority Element -List

Score: 10.00 TC: 1/1 passed

CODE

```
def majority(n):
    for i in n:
        if n.count(i)>len(n)//2:
            return i
    return -1
n=[2,2,1,2,3,2,2]
print(majority(n))
```

INPUT

2 2 1 2 3 2 2

OUTPUT

2

Q6. Sum & Average - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
t=(10,20,30,40,50)
s = sum(t)
avg=s/len(t)
print(s)
print(avg)
```

INPUT

10 20 30 40 50

OUTPUT

150
30.0**Q7. Element Exists -Tuple**

Score: 0.00 TC: 0/2 passed

CODE

```
elements=(10,20,30,40,50)
search=30
if search in elements:
    print("Element Found")
else:
    print("Element Not Found")
```

INPUT

10 20 30 40 50
35

OUTPUT

Element Found

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Q8. Maximum Element - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
a = (12,45,8,67,23)
print(max(a))
```

OUTPUT

67

Q9. Lucky Number

Score: 5.00 TC: 1/2 passed

CODE

```
def lucky(n):
    if n<10:
        return n == 7
    def s(x):
        if x == 0:
            return 0
        return x%10 + s(x//10)
    return lucky(s(n))
n = 34
if lucky(n):
    print("Lucky Number")
else:
    print("Not a Lucky Number")
```

INPUT

123

OUTPUT

Not a Lucky Number

Q10. Remove Duplicate Elements - Tuple

Score: 0.00 TC: 0/1 passed

CODE

```
t=(10,20,30,40,10,50)
result =()
for i in t:
    if i not in result:
        result += (i,)
print(result)
```

OUTPUT

(10, 20, 30, 40, 50)